



Indian Institute of Technology Gandhinagar

MA 104: ODE

Semester-II, Academic Year: 2025-26

Tutorial – 1

- (1) **Formation of Differential Equations.** Formulate the differential equations represented by the following functions by eliminating the arbitrary constants a, b, c :

(a) $y = ax^2$

(b) $y - a^2 = a(x - b)^2$

(c) $y = a(1 - x^2) + bx + cx^3$

(d) $y = cx + f(c)$

- (2) **General Solutions.** Find the general solution of the following differential equations:

(a) $y' = \frac{1}{(x+1)(x^2+1)}$

(b) $x^2y' = y^2 + xy + x^2$

- (3) **Homogeneous Equations.** Verify that the following equations are homogeneous and solve them. (*Recall that $M(x, y)dx + N(x, y)dy = 0$ is said to be homogeneous if M and N are homogeneous functions of the same degree.*)

(a) $(x^2 - 2y^2)dx + xy dy = 0$

(b) $x^2y' - 3xy - 2y^2 = 0$

(c) $x^2y' = 3(x^2 + y^2)\tan^{-1}\left(\frac{y}{x}\right) + xy$

(d) $x^2y' = y^2 + 2xy$

(e) $(x^3 + y^3)dx - xy^2 dy = 0$

- (4) **Singular Solution.** An ODE may sometimes have an additional solution that cannot be obtained from the general solution and is called a *singular solution*. The ODE $(y')^2 - xy' + y = 0$ is of this kind. Show by differentiation and substitution that it has the general solution $y = cx - c^2$ and the singular solution $y = \frac{x^2}{4}$.

- (5) **Change of Variables.** Show that the substitution $z = ax + by + c$ transforms

$$y' = f(ax + by + c)$$

into an equation with separable variables, and apply this method to solve:

(a) $y' = (x + y)^2$

(b) $y' = \sin^2(x - y + 1)$

- (6) **Orthogonal Families.** Let a family of curves be integral curves of $y' = f(x, y)$. Let a second family have the property that at each point $P = (x, y)$ the angle from the curve of the first family through P to the curve of the second family through P is α . Show that the curves of the second family satisfy

$$y' = \frac{f(x, y) + \tan \alpha}{1 - f(x, y) \tan \alpha}.$$

- (7) **Exact Equations.** Determine which of the following equations are exact, and solve those that are:

- (a) $(\sin x \tan y + 1) dx + \cos x \sec^2 y dy = 0$
- (b) $(y - x^3) dx + (x + y^3) dy = 0$
- (c) $(y + y \cos xy) dx + (x + x \cos xy) dy = 0$
- (d) $(e^{y^2} - \csc y \csc^2 x) dx + (2xye^{y^2} - \csc y \cot y \cot x) dy = 0$
- (e) $\frac{x}{(x^2 + y^2)^{3/2}} dx + \frac{y}{(x^2 + y^2)^{3/2}} dy = 0$
- (f) $(2xy^2 + 4)dx = 2(3 - x^2y) dy$

- (8) **Value of n for Exactness.** Find the value of n for which each equation is exact and solve.

- (a) $(xy^2 + nx^2y) dx + (x^3 + x^2y) dy = 0$
- (b) $(x + ye^{2xy}) dx + nxe^{2xy} dy = 0$

- (9) **Integrating Factors.** Solve the following equations using suitable integrating factors:

- (a) $e^x dx + (e^x \cot y + 2y \csc y) dy = 0$
- (b) $(x + 2) \sin y dx + x \cos y dy = 0$
- (c) $y dx + (x - 2x^2y^3) dy = 0$
- (d) $(y \log y - 2xy) dx + (x + y) dy = 0$
- (e) $(x^2y^2 + y) dx + (2x^3y - x) dy = 0$
- (f) $(x^2 + xy) dy + (3xy + y^2) dx = 0$
- (g) $y dx - x dy - (x^2 + y^2) dx = 0$

- (10) **Curved Mirror.** Find the shape of a curved mirror such that light from a source at the origin is reflected in a beam of rays parallel to the x -axis.

- (11) **Substitution Method.** Solve the equation by the substitution $z = y/x^n$ or $y = x^nz$ for a suitable n :

$$\frac{dy}{dx} = \frac{2y}{x} + \frac{x^3}{y} + x \tan\left(\frac{y}{x^2}\right).$$